RNS INSTITUTE OF TECHNOLOGY

Autonomous Institution Affiliated to VTU
RR Nagar, Bengaluru-560098

Estd : 2001

I Semester B.E. Degree Examination – Jan / Feb 2026
Quantum Physics and Applications (CSE Stream)

Time: 3 hrs **Max. Marks: 100**

Note:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***M: Marks, L: Bloom's level, C: Course Outcomes***

Q.No.		Module-1	M	L	C
Q1	a	Explain Einstein's coefficients and derive an expression for energy density of laser system in terms of Einstein's coefficients	8	L2	CO1
	b	Prove that Planck's equation explains the full black body radiation spectrum	8	L2	CO1
	c	An optical fiber has core refractive index 1.5 and clad refractive index 3% less than that of core. Calculate numerical aperture (NA) and angle of Acceptance.	4	L3	CO1
		OR			
Q2	a	Describe the classification of optical fibers with suitable diagrams	8	L2	CO1
	b	Explain construction and working of GaAs semiconductor laser with suitable diagrams.	8	L2	CO1
	c	The average output power of laser source emitting beam of wavelength 632.8 nm is 5mw. Find the number of photons emitted per second by the laser source.	4	L3	CO1
		Module-2			
Q3	a	Apply Schrodinger wave equation to particle in a 1D box and derive expressions for Eigen value and Eigen function	8	L2	CO2
	b	Define group and phase velocity and derive the relation between them	8	L2	CO2
	c	An electron is bound in one dimensional potential well of width 0.18 nm. Find its energy value in eV, for the second excited state.	4	L3	CO2
		OR			
Q4	a	Derive one-dimensional time independent Schrodinger wave equation	8	L2	CO2
	b	State Heisenberg's uncertainty Principle and hence prove that an electron does not exist inside the nucleus.	8	L2	CO2
	c	An electron is associated with a de-Broglie wavelength of 1nm. Calculate the energy and corresponding momentum of the electron.	4	L3	CO2
		Module-3			
Q5	a	What is Qubit? Mentions its properties and describe the representation of a qubit using Bloch's sphere.	8	L2	CO3
	b	Give the matrix representation for Pauli's matrices. Discuss the operation of Pauli matrices on $\|0\rangle$ and $\|1\rangle$ states	8	L2	CO3
	c	Show that the following matrix A is unitary, $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	4	L3	CO3

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		OR			
Q6	**a**	Give the matrix representation of CNOT gate and discuss their operation on four different input states. Also give its truth table.	8	L2	CO3
	b	State Moore's law and elucidate the differences between classical and quantum computing	8	L2	CO3
	c	Given $\|\psi\rangle=\begin{bmatrix}\alpha_1\\ \alpha_2\end{bmatrix}$ and $\|\phi\rangle=\begin{bmatrix}\beta_1\\ \beta_2\end{bmatrix}$ prove that $\langle\psi\|\phi\rangle = \langle\phi\|\psi\rangle^*$	4	L3	CO3
		Module-4			
Q7	**a**	Define Fermi energy and derive an expression for Fermi energy at zero kelvin.	8	L2	CO4
	b	Derive London's equation for superconductors.	8	L2	CO4
	c	Find the temperature of which there is 1% probability that a state with energy 0.5eV above the fermi energy is occupied.	4	L3	CO4
		OR			
Q8	**a**	Describe Type I and Type II superconductors. Give a brief account of high temperature superconductivity.	8	L2	CO4
	b	Describe fermi level in intrinsic semiconductor and hence obtain the expression for fermi level in terms of energy gap of intrinsic semiconductor.	8	L2	CO4
	c	At 6 K, the critical field is $5x10^3$ Am^{-1}. Calculate the transition temperature when critical magnetic field is $2x10^4$ Am^{-1} at 0 K.	4	L3	CO4
		Module-5			
Q9	**a**	Derive the relation between Skip distance and Maximum Usable Frequency for flat surface	8	L2	CO5
	b	Explain the types of remote sensing. Mention its advantages and limitations.	8	L2	CO5
	c	A Radar operates at a frequency 5GHz. A Target is moving towards the RADAR with a speed 150m/s. Find Doppler frequency shift.	4	L3	CO5
		OR			
Q10	**a**	Describe the three methods of propagation of communication waves.	8	L2	CO5
	b	What is Doppler effect in RADAR? derive expression for Doppler frequency.	8	L2	CO5
	c	For an ionospheric layer at height of 300km with an electron density of $5x10^{11}$ electrons/m^3, and skip distance of 1039 km, Calculate the critical frequency and maximum usable frequency.	4	L3	CO5

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CIE – Test 1

Sem: I
Subject: Quantum Physics and Applications for CSE stream
Subject Code: BAPCS102

Date:14/11/2025
Time: 9:30-11:00 am
Max. Marks: 50

Instructions to the students:

1. ***Answer any 5 full questions, selecting at least one question from each module.***
2. ***Missing data may be suitably assumed.***
3. ***Physical constants: $c = 3 \times 10^8$ m/s; $k = 1.38 \times 10^{-23}$ J/K; $h = 6.625 \times 10^{-34}$ Js; $\varepsilon_o = 8.854 \times 10^{-12}$ F/m; $me = 9.1 \times 10^{-31}$ kg.***

Q.No.		Question	Marks	RBT*	CO
1	a	Deduce Wein's Law and Raleigh Jeans Law from Planck's law.	6	L2	CO
	b	The ratio of population of two energy levels is 1.059×10^{-30}. Find the wavelength of light emitted at 330K.	4	L3	
		OR			
2	a	Obtain an expression for energy density in terms of Einstein's coefficients	6	L2	CO
	b	In an optical fiber core diameter is found to be 50 µm, core and cladding refractive indices 1.45 and 1.40 respectively where wavelength is 820 nm. Find numerical aperture and number of modes of propagation.	4	L3	
3	a	Derive an expression for numerical aperture of an optical fiber. Mention the condition for ray propagation.	6	L2	CO
	b	Explain the conditions for laser action.	4	L2	
		OR			
4	a	Explain different types of optical fibers with neat diagram.	6	L2	CO
	b	Explain with block diagram the point-to-point communication	4	L2	
5	a	State de-Broglie's hypothesis and derive an expression for the de-Broglie wavelength in terms of accelerating potential.	6	L2	CO
	b	A spectral line of wavelength 4000 Å has a width of 8×10^{-5} Å. Evaluate the minimum time spent by the electrons in the upper energy state.	4	L3	
		OR			
6	a	State and Explain Heisenberg's Uncertainty Principle and hence prove that an electron does not exist inside the nucleus.	6	L2	CO
	b	Calculate the de-Broglie wavelength associated with an electron of energy 100 eV.	4	L3	
7	a	Set up Schrodinger's time-independent wave equation in one dimension.	6	L2	CO
	b	An electron is bound in an infinite potential well of width 0.2 nm. Find its energy values in the first two allowed energy states.	4	L3	

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		OR			
8	a	Apply Schrodinger's time-independent wave equation to arrive at the normalized wave function for a particle in a 1-D box.	6	L2	CO2
	b	Calculate the de-Broglie wavelength of a particle of mass 0.65 MeV/c^2 having a kinetic energy 80 eV.	4	L3	
9	a	Elucidate the differences between Classical and Quantum computing.	6	L2	CO3
	b	Check if, matrix A=$\begin{bmatrix} 1 & 4+3i \\ 4-3i & 5 \end{bmatrix}$ is Hermitian.	4	L3	
		OR			
10	a	Explain the matrix representation of 0 and 1 states and applyidentity operator I to $\|0\rangle$ and$\|1\rangle$ states.	6	L2	CO3
	b	Prove that, A=$\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & \frac{-i}{\sqrt{2}} \end{bmatrix}$ is Unitary.	4	L3	

** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create*

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CIE TEST II

m: I

bject: Quantum physics and Applications (CSE Stream) **Date: 27-12-2025**

bject Code: BAPCS102 **Time: 9.30 – 11.00 a.m**

Max. Marks: 50

sical constants: *Planck's constant* $h=6.634\times10^{-34}$Js, *mass of electron* $m=9.1\times10^{-31}$kg, *ctron charge* $e=1.6\times10^{-19}$C, *Velocity of light* $c=3\times10^{8}$m/s *Boltzmann's constant* $k=1.38\times10^{-23}$ J/K

tructions: Answer any 5 full questions choosing One full question from each choice

Q. N.		Questions	Marks	RBT*	COs
	a	Give the matrix representation of CNOT gate and discuss their operation on four different input states	6	L2	CO3
	b	Show that S gate can be formed by connecting 2 T gates in series	4	L3	CO3
		OR			
	a	Write Pauli's matrices and operate Pauli's matrices on the states $\|0\rangle$ and $\|1\rangle$	6	L2	CO3
	b	The Hadamard gate is represented by the matrix $H=\frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}$ Show that H is unitary	4	L3	CO3
	a	Describe fermi level and derive an expression for Fermi Energy at 0K.	6	L2	CO4
	b	The free electron concentration in copper is $84.5\times10^{27}/m^3$. Calculate the Fermi energy and Fermi temperature.	4	L3	CO4
		OR			
	a	Derive London's equation for superconductors.	6	L2	CO4
	b	Find the temperature at which there is 1% probability that a state with energy 0.5eV above fermi energy is occupied	4	L3	CO4
	a	Mention the expressions for electron and hole concentrations and prove that Fermil evel in intrinsic semiconductor lies in the middle of the band gap.	6	L2	CO4
	b	At 6K Critical field is 5×10^{3} Am^{-1}. Calculate the transition temperature if the critical magnetic field at 0K is 10×10^{3} Am^{-1}	4	L3	CO4
		OR			

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6	a	Explain Meisner effect and describe Type I and Type II superconductors	6	L2	CO4
	b	In a metal of Fermi energy 0.38eV calculate the probability of an electron occupying an energy level 0.4eV at 300k.	4	L3	CO4
7	a	Derive the relation between Skip distance and Minimum Usable Frequency for flat surface	6	L2	CO4
	b	With a transmit frequency of 5GHz Calculate doppler frequency seen by stationary radar when the target radial velocity is 100km/h	4	L3	CO4
		OR			
8	a	Explain the types of remote sensing. What are the advantages and limitations?	6	L2	CO5
	b	A radio wave is incident on a layer of the ionosphere at an angle of 30° to the vertical. If the critical frequency is 1.2 MHz and the ionospheric height is 300 km, calculate: (i) Maximum usable frequency (MUF) (ii) Skip distance	4	L3	CO5
9	a	Describe the three methods of propagation of communication waves	6	L2	CO5
	b	The operating frequency of radar is 3GHz if the relative velocity of target is 500kmph, find out the doppler shift	4	L3	CO5
		OR			
10	a	What is Doppler effect in RADAR? derive expression for Doppler frequency	6	L2	CO5
	b	For an ionospheric layer at height of 300km with an electron density of $5x10^{11}$electons/m^3, and skip distance of 1039 km, Calculate the critical frequency and maximum usable frequency.	4	L3	CO5

** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create*

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CIE – Test 1

n: II
ject: Applied Chemistry for Smart Systems (CSE Stream)
ject Code: BACCS202

Time: 9:30-11:00
Max. Marks: 50
Date: 13/04/2026

ruction: Answer any 5 full questions, selecting ONE question from each part.

No.		Question	Marks	RBT*	COs
	a	Define photovoltaic cell. Explain the construction and working of a photovoltaic cell.	6	L2	CO1
	b	Elaborate the components of Na-ion battery and explain how electric energy is produced.	4	L2	
		OR			
	a	Discuss the construction and working of ultrasmall asymmetric supercapacitors.	6	L2	CO1
	b	Explain how fuel cells differ from conventional cells.	4	L2	
	a	With a neat labelled diagram discuss the construction and working of solid-oxide fuel cell.	6	L2	CO1
	b	A concentration cell was constructed by immersing two silver electrodes in 0.03M and 3M $AgNO_3$ solutions. Write the cell representation, cell reactions and calculate the emf of the cell at 25^0C.	4	L3	
		OR			
	a	Outline the construction and working of Vanadium redox flow battery.	6	L2	CO1
	b	The emf of the cell Cu| $CuSO_4$(0.001 M) ‖$CuSO_4$(x M) | Cu is 0.0595V at 30^0C. Find the value of x.	4	L3	
	a	Explain the electrochemical theory of corrosion by taking iron as an example.	6	L2	CO2
	b	Calculate the CPR in both mpy and mmpy for a thick steel sheet of area 200 in^2 which experiences a weight loss of 845g after 3 year. (Density of steel=7.9g/cm^3).	4	L3	
		OR			
	a	Describe the application of a conductometric sensor in determining an acid mixture.	6	L2	CO2
	b	Determine the area of the metal, if its density 15 g/cm^3 has lost 10.5 kg in 4 years at a CPR 395mpy.	4	L3	
	a	What is anodizing? Explain the process of anodizing of aluminium with diagram and relevant reactions.	6	L2	CO2
	b	Illustrate how vapour corrosion inhibitors are able to protect computer circuit boards.	4	L2	
		OR			

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8	a	Explain the application of electrochemical sensor in detection of SO_x and NO_x	6	L2	CO2
	b	A steel screw is surrounded by a copper rivet and exposed to a moist environment. Explain which metal undergoes corrosion and justify your answer with reactions.	4	L2	
9	a	Illustrate green synthesis of ZnO nanoparticles by solution combustion method with reactions.	6	L2	CO3
	b	List out any three properties and two applications of green solvents for server heat management.	4	L2	
		OR			
10	a	Outline the synthesis of glycerol trioleate ester. Describe its properties and applications	6	L2	CO3
	b	Discuss any 4 principles of green chemistry.	4	L2	

** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create*

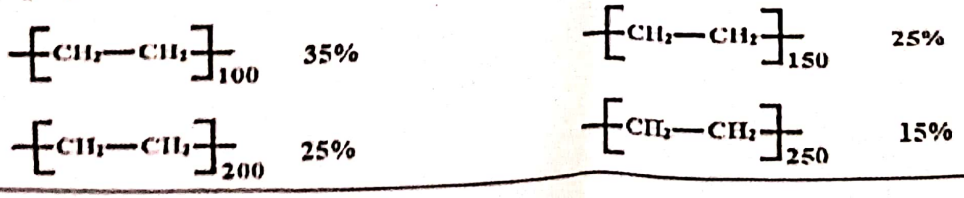
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CIE – Test II

Sem: II
Subject: Applied Chemistry for Smart Systems (CS Stream)
Subject Code: BACCS202

Date: 01-06-2026
Time: 9:30 - 11:00am
Max. Marks: 50

Instruction: Answer any 5 full questions, selecting ONE question from each part.

Q. No.		Question	Marks	RBT*	COs
1	a	Explain the extraction of gold from e-waste by aqua regia method	6	L2	CO3
	b	Outline the synthesis of PLA. Mention any two properties and applications of PLA.	4	L2	
		OR			
2	a	Outline the hydrometallurgical extraction of copper from e-waste.	6	L2	CO3
	b	Mention the effects of e-waste on human health and environment	4	L2	
3	a	Explain the construction and working of QDSSC with relevant reactions	6	L2	CO4
	b	Explain the structure property relationship of polymers with respect to (i) Crystallinity (ii) Plastic deformation	4	L2	
		OR			
4	a	Describe the synthesis of Cd-Se quantum dots by wet chemical method with suitable chemical reactions	6	L2	CO4
	b	Give the synthesis of PMMA. List its applications.	4	L2	
5	a	Give the synthesis of polyaniline and explain its conduction mechanism	6	L3	CO4
	b	Write a note on size dependent properties of quantum dots	4	L2	
		OR			
6	a	Calculate the number average and weight average molecular weight of a polymer with the following composition 	6	L3	CO4
	b	Explain the synthesis of nylon-12. Mention its applications	4	L2	

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7	a	Discuss the synthesis of TiO_2 nanoparticles by sol-gel method for ReRAM applications.	6	L2	CO5
	b	Differentiate between organic and inorganic memory devices.	4	L2	
		OR			
8	a	Describe the construction and working of pentacene semiconductor chips with a schematic diagram	6	L2	CO5
	b	Mention any four properties and applications of TiO_2 nanoparticles	4	L2	
9	a	What are liquid crystals? Briefly explain the classification of thermotropic liquid crystals with examples.	6	L1, L2	CO5
	b	Outline the working principle of LED	4	L2	
		OR			
10	a	Illustrate the construction and working of Active Matrix OLED	6	L2	CO5
	b	List the properties and applications of QLED.	4	L2	

*** Revised Bloom's Taxonomy: L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create**

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RNS INSTITUTE OF TECHNOLOGY

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I / II Semester B.E. Degree Examination – June / July 2026
Applied Chemistry for Smart Systems (CSE stream)

Time: 3 hrs **Max. Marks: 100**

Note:
1. *Answer any 5 full questions, selecting at least one question from each module.*
2. *Missing data may be suitably assumed.*
3. *M: Marks, L: Bloom's level, C: Course Outcomes*

Q.No.		Module-1	M	L	C
Q1	a	Define battery. Explain the classifications of battery with suitable examples.	7	L1, L2	CO1
	b	Summarize the construction, working and applications of Lithium-ion battery with a neat, labelled diagram.	7	L2	CO1
	c	Mention any six differences between battery and fuel cells.	6	L2	CO1
		OR			
Q2	a	Explain the construction, working and applications of ultra-small asymmetric super capacitors with a schematic diagram.	7	L2	CO1
	b	Describe the construction, working and applications of solar photovoltaic cell with a neat, labelled diagram.	7	L2	CO1
	c	A cell is constructed by immersing two silver electrodes in 0.01 M and 1.0 M $AgNO_3$ solutions. Write the cell reaction, cell representation and calculate the EMF of the concentration cell.	6	L3	CO1
		Module-2			
Q3	a	Summarize the electrochemical theory of corrosion by taking iron as an example.	7	L2	CO2
	b	Briefly discuss the working principle and applications of electrochemical sensor in the detection of SOx and NOx with a neat, labelled diagram.	7	L2	CO2
	c	A steel area 100 $inch^2$ is exposed to air near the seashore. After 1 year it was found that the steel sheet has lost 485g due to corrosion. What is the value of CPR in mils/year and in mm/year? (Given density of steel = 7.9 g/cm^3)	6	L3	CO2
		OR			
Q4	a	Discuss the estimation of copper present in electroplating effluent through colorimetric pathway.	7	L2	CO2
	b	A thick steel sheet of area 400 $inch^2$ is exposed to air near the ocean. After two-year period it was found to experience a weight loss of 375g due to corrosion. What is the value of CPR in mils/year and in mm/year? (Given density of steel = 7.9 g/cm^3).	7	L3	CO2
	c	Describe galvanization process with a neat, labelled diagram.	6	L2	CO2

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		Module-3			
Q5	a	Explain synthesis, properties and applications of glycerol trioleate.	7	L2	CO3
	b	Summarize the steps involved in the extraction of copper from e-waste by hydrometallurgical process.	7	L2	CO3
	c	Discuss the synthesis of ZnO nanoparticles under green chemistry strategy.	6	L2	CO3
		OR			
Q6	a	Give the properties (any four) and applications (any three) of the green solvent for sever heat management.	7	L2	CO3
	b	Explain the effect of E-waste on environment and human health.	7	L2	CO3
	c	Enumerate the synthesis and applications of Polylactic acid (PLA).	6	L2	CO3
		Module-4			
Q7	a	Discuss the following structure property relationship of polymers: (i) Branching & Cross linking (ii) Crystallinity	7	L1	CO4
	b	Explain construction, working and applications of Quantum dot sensitized solar cells (QDSSCs) with a neat, labelled diagram.	7	L2	CO4
	c	Explain synthesis of PMMA and provide its major properties (any four).	6	L2	CO4
		OR			
Q8	a	A polymer sample containing 50, 100 and 150 molecules having molar mass of 1000 g/mol, 2000 g/mol and 3000 g/mol respectively. Calculate number average and weight average molecular mass of the polymer.	7	L3	CO4
	b	What are conducting polymers? Explain the mechanism of conduction in Polyaniline.	7	L1, L2	CO4
	c	Explain the size dependent properties of quantum dots with respect to band-gap variation and Quantum confinement effect.	6	L2	CO4
		Module-5			
Q9	a	Explain the role of pentacene used in p-type memory devices.	7	L2	CO5
	b	Discuss construction, working and applications of OLEDs with a neat, labelled diagram.	7	L2	CO5
	c	Define liquid crystals. Explain its classifications in broad with examples	6	L1,L2	CO5
		OR			
Q10	a	Illustrate the construction, working and applications of QLEDs.	7	L2	CO5
	b	Discuss the process of synthesizing TiO_2 nanomaterials employed in Resistive RAM (ReRAM) applications.	7	L2	CO5
	c	Mention the differences (any six) between organic and inorganic memory devices.	6	L1	CO5

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